

User Manual (Step-by-Step)

1. Introduction

This system allows users to report and track service outages such as electricity, water, and internet.

2. System Requirements

- Node.js (v16+)
- Browser (Chrome recommended)
- Internet connection

3. Installation

1. Open project folder
2. Run backend:
cd backend
npm install
node server.js
3. Run frontend:
cd frontend
npm install
npm run dev

4.1 Login / Registration

1. Open application
2. Click **Login / Register**
3. Enter:
 - Name
 - Email
 - Password
4. Select role (default = customer)
5. Click **Submit**

4.2 Reporting an Outage (Customer)

1. Go to **Report Outage Page**
2. Fill:
 - Title
 - Description
 - Location
 - Type (electricity/internet/water)
3. Select location (latitude & longitude)
4. Submit

Outage will be marked as **Pending**

4.3 Viewing Outages

- Navigate to **Outage List Page**
- Filter by:
 - Status
 - Type
 - Priority

4.4 Admin Assigning Outage

1. Go to **Admin Dashboard**
2. Select outage
3. Assign technician
4. Update priority/status

4.5 Technician Workflow

1. Open **Assigned Outages Page**
2. Select outage

3. Update:

- Status → In Progress / Resolved
- Add resolution notes

4.6 Notifications

- Users receive updates when:
 - Outage assigned
 - Status changed
- View in **Notifications Page**

4.7 Dashboard

Each role sees:

- Total outages
- Pending vs resolved
- Assigned tasks (for technician)